



Global Airborne Observatory

A visual approach to bridge the gap between science and decision making to conserve the environment. <http://dgit.in/mar21-17>



Forest carbon content

A new model for measuring forest carbon content from space, through remote sensing and machine learning. <http://dgit.in/mar21-18>

Technology vs deforestation

How drones and satellites are helping conserve forest cover

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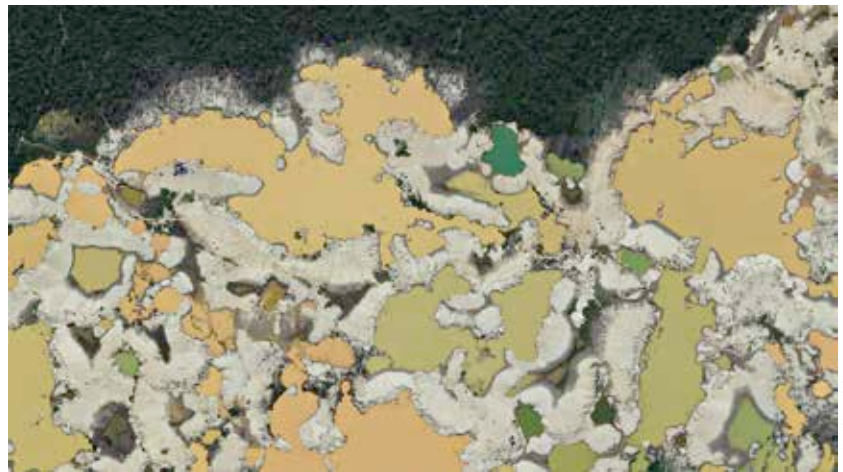
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istorically, combating deforestation has been problematic because of how remote the locations

are, and the vastness of the areas that need to be monitored. It is simply not possible for humans to keep track of all the world's forests. However, sophisticated new technologies can monitor large swathes of land, and even the whole planet, produce high resolution 3D maps of vegetation, investigate the chemical properties of the forest cover and provide a whole range of unprecedented insights, that are all helping conserve forests, which are the lungs of the Earth.

SATELLITE ALERTS

The Global Land Analysis and Discovery system is a service launched in 2016 that allows interested parties to subscribe to alerts based on satellite imagery, which pings whenever regions of deforestation are detected. Computers compare current latest remote sensing data to historical Landsat data from the early 80s to detect the regions where deforestation is occurring. Governments, NGOs, and other organisations interested in tracking forest cover can subscribe to these alerts for free. In 2021, a study into the impact



Tailings or excavated depressions produced by informal gold mining in Peru

of GLAD found that two African countries who subscribed to the alerts managed to reduce deforestation by 18 percent. The countries that were simply just covered by GLAD and did not have official bodies or organisations subscribe to the satellite alert service, did not get a boost in combating deforestation. However, the same study showed a discrepancy in Asia and South America, where even the countries that subscribe to the alerts did not benefit from them. The reason identified by the researchers was that in the Asian countries monitored, there were already conservation efforts that depended on satellite based alerts, and political unrest in South America. Interested organisations and parties can go through a wealth of data available on Global Forest Watch, and subscribe to global alerts or choose an area to monitor and get alerts on from that region. The alerts can be delivered as frequently as once every 8 days, if the conditions for observations are clear.

Satellite intelligence started off as an agritech startup, delivering environmental degradation alerts to food and agricultural industries to optimise their supply chains. Sat Intelligence helps growers, manufacturers and financial institutions track deforestation, fire risk, drought and flooding. Satellite imagery, supply chain data, machine learning algorithms and human intelligence are all part of the system. The system uses data gathered from radar satellites, such as the Sentinel 1, which "sees" the Earth very differently from optical satellites such as Sentinel 2 or Landsat. The radar data for a single pixel can be extracted and charted over time. The backscatter from forests is stronger than the backscatter from deforested areas, and this allows researchers to identify the areas on the world where deforestation has occurred. Algorithms doing the heavy lifting and reading the radar data can be configured to make guesses and predictions on areas where deforestation